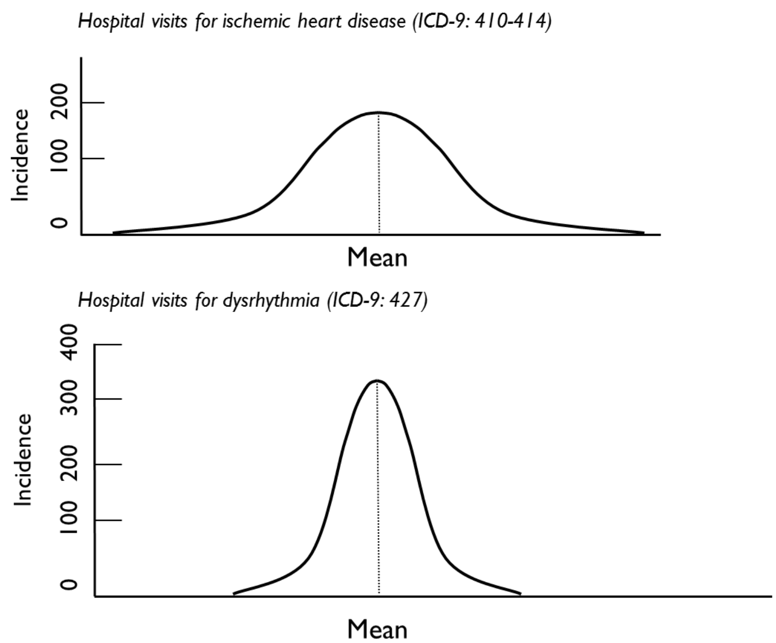
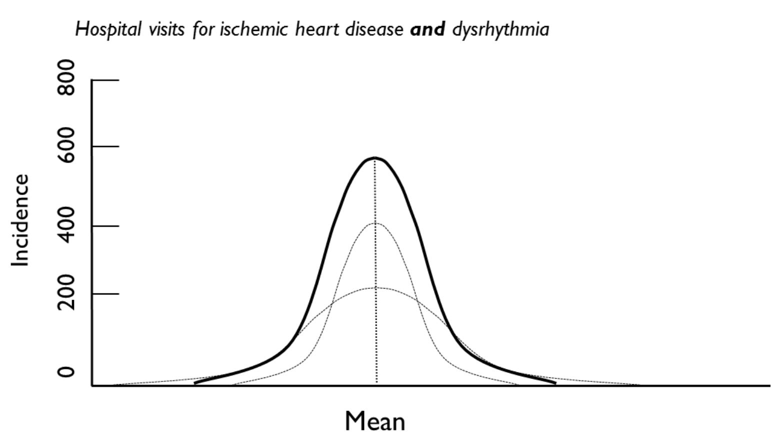
Example 1: Pooling by Addition

You might want to use the *Pooling by Addition* option if you would like to aggregate two outcomes that are non-overlapping. In the example below, we have estimated ischemic heart disease hospital admissions and dysrhythmia hospital admissions. You'll see that each endpoint is associated with unique, and non-overlapping, International Classification of Disease 9th edition (ICD-9) code (Slee 1978). Therefore, it's ok to add the two estimates, because doing so would not double-count impacts.



Addition allows us to combine non-overlapping estimates of a common health endpoint

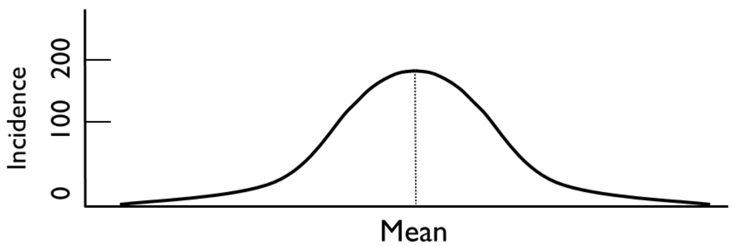


The sum of ischemic heart disease and dysrhythmia is provides a better overall characterization of the effects of air pollution on cardiovascular outcomes than either endpoint alone.

Example 2: Pooling by Subtraction

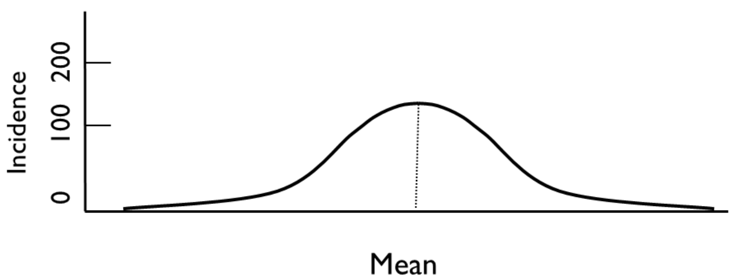
In this example, suppose that you have estimated the incidence of total cardiovascular hospital admissions using one health impact function. Suppose also that you estimated the incidence of total cardiovascular hospital admissions, *less stroke*. In this instance, you could subtract the second incidence estimate from the first incidence estimate to yield the number of stroke hospital admissions.

Hospital visits all cardiovascular outcomes (ICD-9: 390-459)



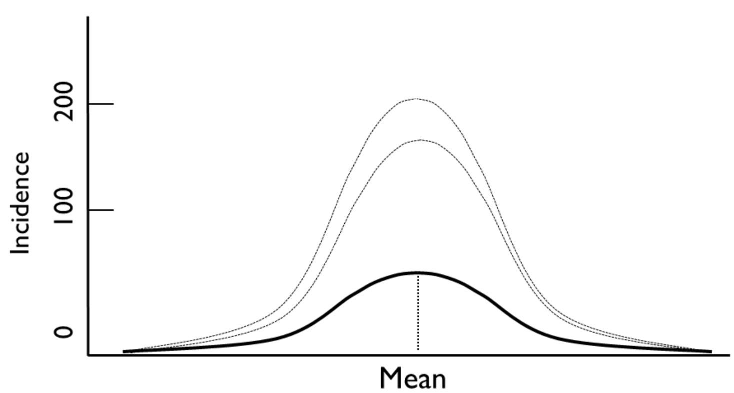
Subtraction allows us to "net out" the incidence of a health endpoint from two or more studies

Hospital visits for all cardiovascular outcomes except stroke (ICD-9 390-440)



In this example, the only difference between these two studies is that study one includes all cardiovascular outcomes, while study two excludes strokes

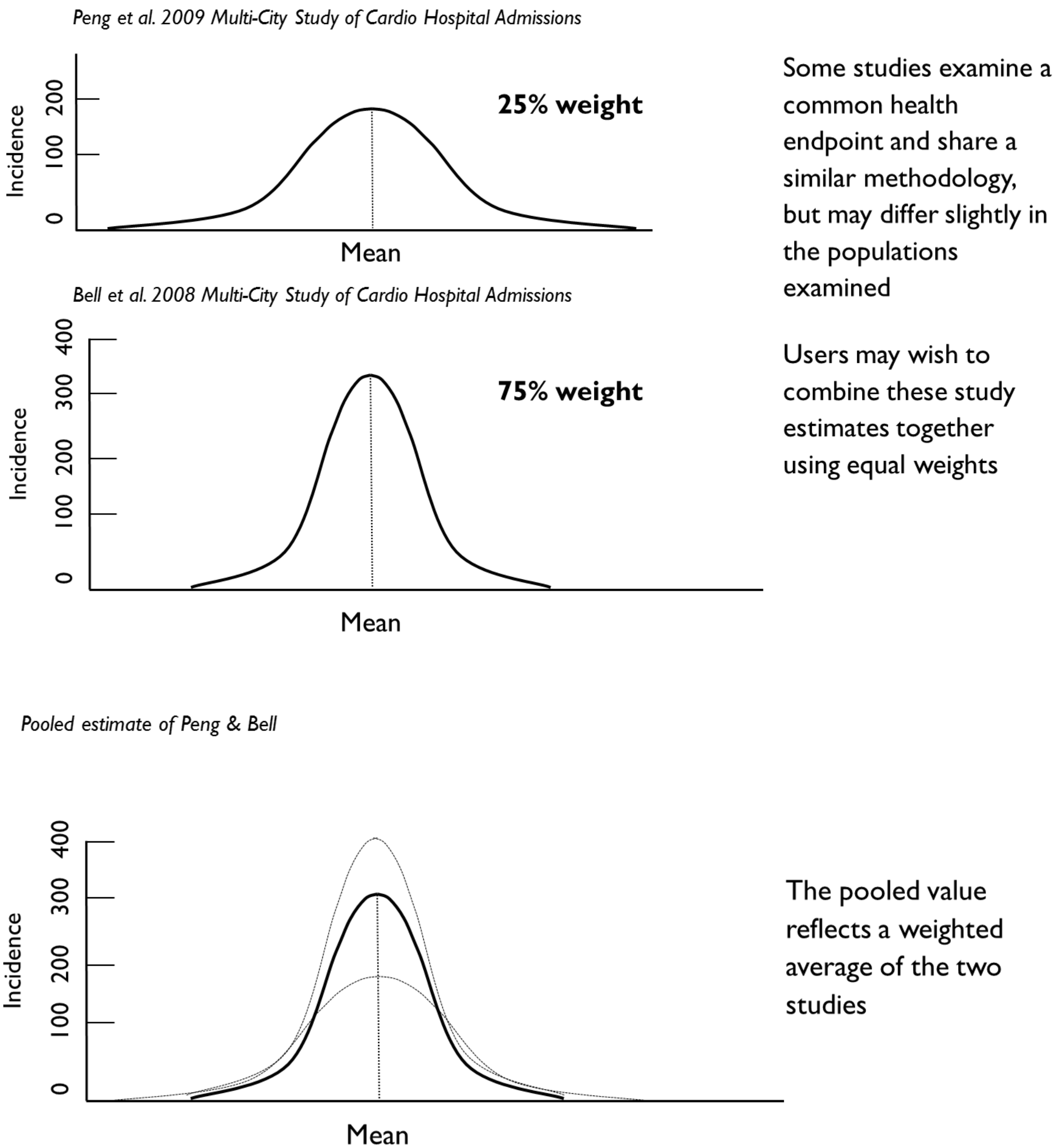
Hospital visits for stroke (ICD-9: 440-449)



Subtracting the results of study two from study one yields an estimate of stroke

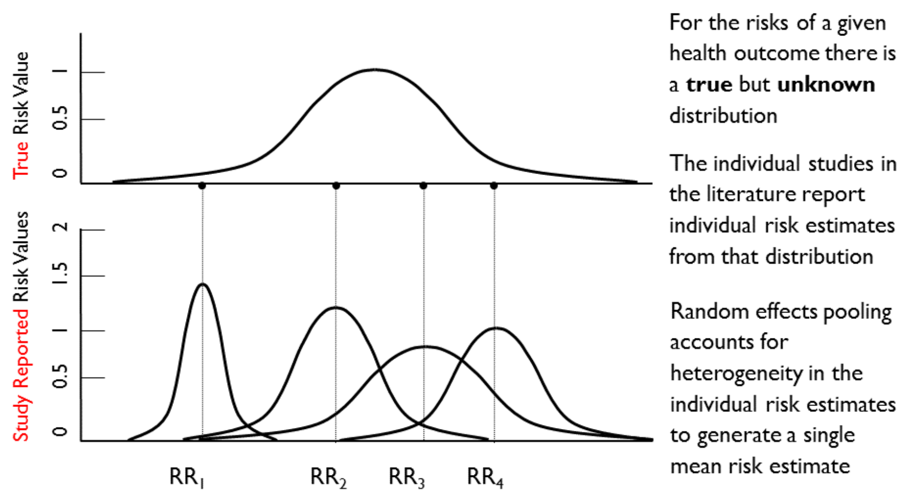
Example 3: Pooling with User-Assigned Weights

In this case, you might have estimated the change in incidence using two different health impact functions for the same health endpoint and would like to combine them using weights that you specify.

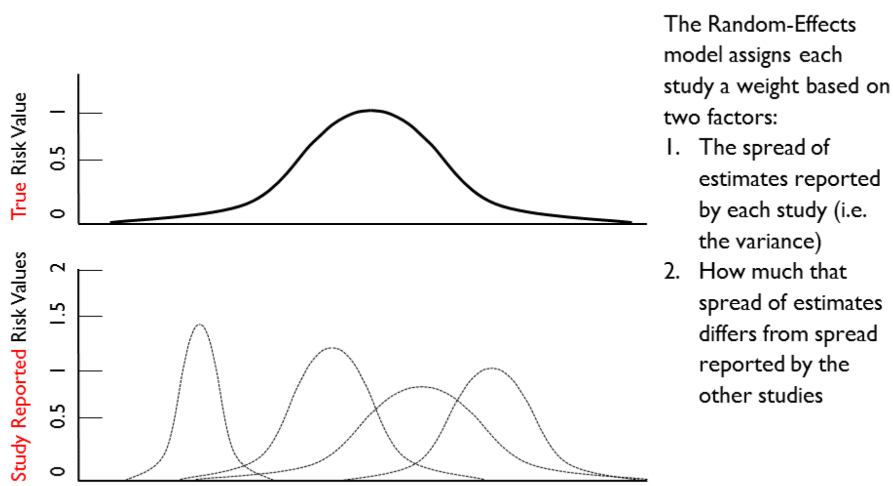


Example 4: Pooling Using the Fixed Effect and Random Effects

The Fixed Effect and Random Effects pooling techniques are among the most complicated and are best applied only when you understand clearly the assumptions inherent in the method and its suitability to the incidence estimates. The example below describes the procedure for performing this technique.



Adapted from: Mosteller and Colditz (1996); Charles Poole EPID 731



Adapted from: Mosteller and Colditz (1996); Charles Poole EPID 731